

Out-of-State Tag Holder Review — one-page summary

Deliverable: `case_predictions.csv` — 24,000 rows (12,000 cases × T0/T1) in the template's exact schema. Full write-up in `METHOD.md`.

The problem behind the problem

The package gives no key linking `candidate_records.csv` to the six evidence feeds — only names, and the names are deliberately corrupted. Getting the identity layer right mattered more than the choice of classifier, so that is where the work went.

Using `vehicle_ref` on the title feed as an independent check, the corruption pattern is **3 good rows plus 1 replaced outright**: cluster the owner names inside each vehicle's four rows and 98% split three-and-one, with the fourth name unrelated to the other three. Grouping the licence feed by date of birth reproduces the same pattern, so it is the generator's noise model, not a quirk of one table.

Two consequences:

- The title feed **can** recover its fourth row, because `vehicle_ref` says which vehicle it belongs to. Doing so lifts that feed from 38,582 to 47,581 linked rows and its signal from $\rho +0.25$ to $+0.37$.
- The other feeds **cannot**. Forcing an assignment there made the signal *worse* — an unrelated name carries an unrelated state. Accepting ~80% recall is the better trade, and 9,369 of 12,000 candidates land on exactly three address rows as predicted.

The classes are ordered, so the model is

In the development labels no case ever jumps from `review_not_warranted` to `review_warranted` between phases — every move passes through `insufficient_evidence`. That is one latent score with two thresholds, so the model is a **proportional-odds ordinal logistic**: one direction vector and two cut-points instead of three independent classifiers, which matters at 300 labelled candidates. On identical features it beats multinomial logistic (0.953), random forest (0.981) and gradient boosting (1.038).

The direction is **recency**: cases whose recent records point to Delaware are the ones warranting review. A 150-day exponential half-life beats every alternative aggregation tried, including a change-point fit. The work-location feed is complete, uncorrupted and correlates -0.07 with the answer — it is excluded from the pooled score.

Results

Grouped 5-fold cross-validation, 20 repeats, 600 labelled rows:

	model	baseline
log loss	0.934	1.096
accuracy	0.527	0.355
macro F1	0.528	0.175
macro AUC	0.719	0.500
priority ranking AUC	0.781	0.500

T1 scores better than T0 (0.914 vs 0.954) — the model moves on the later evidence. Probabilities are calibrated with no post-hoc fitting (predicted vs observed agree within ± 0.05 across quintiles). Errors sit on the adjacent class: only 39 of 600 rows cross the whole scale.

$\text{review_priority} = p(\text{warranted}) + 0.5 \times p(\text{insufficient})$ — a case that warrants review takes full priority, a case the evidence cannot decide still needs a person and takes half.

Honest ceiling

Eight further ideas were tested and rejected, including semi-supervised self-training (worse at every setting) and three routes to recovering the unattributable rows. Each candidate has roughly fourteen state-bearing records split between two states, so the latent direction can only be estimated so precisely. Macro AUC ~ 0.72 looks close to what this data supports.

For the reviewer

`case_audit.csv` carries the same 24,000 rows plus the evidence behind each score — newest record per feed with its date, evidence freshness, T1 update direction, and the signed contribution of each feature block — so a case can be checked without reading the code. `diagnostics.py` and `model_selection.py` reproduce every number quoted above.

The output supports staff review. It is not a residency, fee, or enforcement determination.